Agentic CMS
Project overview

One agency workspace that runs the whole blog programme for many client websites: the agency approves ideas, and agents draft, illustrate, check and render each post into the client's own template, open the merge request and publish it.

STATUS	APP	HARNESS	DATE
Frontend prototype complete	apps/cms · @studio/cms	@studio/seo-harness 0.2.0	5 November 2026

01 · SUMMARY

What this is

Agencies run SEO blogs for many clients. Today each post means a writer, a designer, an SEO check and a developer pasting HTML into the client's codebase. The Agentic CMS removes everything except the decisions that need a person.

1
human decision per post by default: approving the idea

11 + 1
harness agents plus the CMS agent that owns the repo

8
client sites seeded in the demo workspace, 7 people

156
automated tests passing; type-check and production build clean

The promise

Run a client's entire blog programme without opening a code editor. Every post lands in the client's own site template, through a normal merge request, with every claim sourced and every action audited.

What's been delivered

- A full working frontend prototype for many sites and many users
- A design system built from the reference screens
- The SEO harness packaged into the monorepo
- A backend blueprint, schema and ten ordered build handoffs

02 · HOW A POST MOVES

Only ideas need approval

Autopost is on by default for every site. A person approves the idea; the agents do the rest, and stop for a person only when a guardrail fires.

Idea
From keyword research, a ranking gap or a person. A specialist approves it.

→

Draft
Writing agent drafts in the site's brand voice, marking every factual claim.

→

Images
Visual agent generates, diagrams or sources images, with provenance.

→

Gate
SEO checks run. Agents fix what they can, and source or rewrite unsourced facts.

→

Merge request
Rendered into the client's template, committed on its own branch.

→

Live
Merges itself when the client's CI is green. Ranking measured after 24 h.

When a post is held it stops at In review with the reason on the ledger. Reasons include a hard check that can't be fixed, no repo, a template that can't render the post, a budget cap, a provider failure, or an image slot marked for a person.

When autopost is off for a site, every post stops at In review for a specialist to approve, and a developer merges. Auto-merge itself is an admin-only setting.

03 · WHO USES IT

Roles are per site, not global

One person can be a specialist on four sites and have no access to the rest. Workspace owners and admins act as admin on every site. Locked actions always explain why instead of disappearing.

ROLE	DOES	CAN'T
Content specialist	Approves ideas and content, edits posts, sends drafts back, overrides soft checks, triggers runs, turns autopost on or off	Merge, connect repos, change budgets
Developer	Connects the repo and Search Console, uploads and confirms blog templates, merges and rebases merge requests	Approve content
Client stakeholder	Sees their own site, its plan, posts and SEO reports; comments	Anything that changes content or settings
Agency admin	Everything, plus budgets, model routing, members, new sites and the auto-merge policy	—

Working on many sites at once

Sites portfolio

Every client in one view, as a grid or a table: status, ideas waiting, gap to projection, spend against cap and who's online.

Inbox

Everything across my sites that needs me: ideas to approve, held posts, image slots for a person, overdue ledger items, failing CI, invites.

Team

People against sites with each person's role, plus invites. Seat use is shown against the workspace plan.

Site in the URL

Every site screen lives under `/s/<site>/…`, so two tabs or two people can work on different clients at the same time.

Presence and edit locks

Avatars show who else is on a site. If someone is editing a post, others see "view only" with a take-over option.

Tenant isolation

A site you have no access to is hidden from the switcher and shows "Request access". Tests check that one client's data never appears on another's.

04 · WHAT'S IN THE APP

Screens

AREA	SCREENS	WHAT IT DOES
Plan	Board · calendar · pillar matrix · new idea	The content plan by week and pillar, with ideas waiting for approval and the projection
Editor	Post · versions · assets	Draft with inline agent suggestions, fact markers, gate checks, comments, version history and image slots
Live preview	Desktop · tablet · mobile	The post rendered into the client's real template, exactly as it will look once published
Images	Asset library · visual agent	Candidates per slot with their source and licence; generate, diagram or source from the web
Templates	Library · upload (.html / .zip)	Developers upload the blog template; the CMS detects its structure and asks to confirm it
Approvals and MRs	Approvals queue · merge requests	What's waiting on a person, and every merge request with its CI status
SEO	Trajectory · KAS · reports	Search Console traffic against projection, keyword attribution scores and biweekly reports
Operations	Runs · settings · onboarding	Agent runs and their cost, budgets and caps, connections (image MCPs and APIs), schedules, site setup
Workspace	Sites · inbox · team · account · sign-in	The multi-site layer described above

05 · DESIGN SYSTEM

Glass hierarchy, adapted from the Nawa template

Hierarchy

Page → canvas → soft colour washes → frosted shell → white cards → small chips. One black primary button per region.

Type

General Sans 500 for titles and UI, **Poppins 400** for body and meta.

Colour means something

Agent blue machine work · Human amber waiting on a person · Red hard block · Green pass.

Rules are enforced by tests: colours only from tokens, layout that works right-to-left, icons through one registry, WCAG contrast checked. The full kit is shown at `/styleguide`; the documents are `docs/design-system.md` and `docs/02-design-system.cms.md`.

06 · THE AGENTS

SEO harness, packaged in the repo

The harness lives in `packages/seo-harness`: each agent has a written skill, typed inputs and outputs, a model tier and a schedule. The CMS agent is the product's own agent and the only thing that writes to a client repo.

AGENT	MODEL TIER	RUNS	PRODUCES
SEO action plan	Opus 5.5	On demand, or when a pivot is triggered	The plan and pillar keywords
Keyword research swarm	Sonnet 5	Weekly, Monday	New keyword pipeline and top 10
Blog writing agent	Sonnet 5 → Opus for pillar pages	Per publish slot	Draft, fact list, image sheet
On-page SEO optimizer	Sonnet 5	After each draft	Gate checks and fixes
Visual asset agent	Sonnet 5	After each draft	Images and diagrams with provenance
Technical SEO monitor	Haiku 4.5	Weekly, Friday	Site health issues
SEO QA agent	Sonnet 5	Weekly light, biweekly full	Ledger items
Content refresh agent	Sonnet 5	Monthly	Refresh briefs for decaying posts
Link building prospector	Sonnet 5	Monthly	Outreach targets (sending needs a person)
Visualization report agent	Sonnet 5 → Opus on pivot	Biweekly, after QA	The client report
Orchestrator	Haiku 4.5	—	Replaced by deterministic scheduling in the backend
CMS agent	Product agent	Every post	Rendering, merge requests, autopost fixes, auto-merge

Model IDs must be checked against Anthropic's docs before deploying. Always a human decision: approving ideas and the plan, sending outreach, deleting pages, spending over a cap.

Images and templates

Bring your own image providers

Each site can connect image-generation MCP servers or APIs, plus stock and open-licence sources (Openverse, Wikimedia, Unsplash, Pexels). Keys stay on the server, encrypted. Every candidate records where it came from and its licence.

Bring your own blog template

Developers upload the template as HTML or a zip. The renderer only fills in the structure it found and never invents markup; if a post needs something the template can't show, the post is held instead of shipped wrong.

07 · WHERE IT STANDS

Status and what's next

Done and checked

- Every planned screen built for desktop and mobile
- Multi-site, multi-user flows checked in the browser: sign-in, portfolio, no-access, redirects from old links, two-tab edit lock and take-over
- Real template upload and rendering, live preview, image connections
- 156 tests pass; type-check clean; production build green

Not real yet

- Everything runs in the browser: data is local and agent work is simulated with timers
- Sign-in is a "pick a person" demo, not Clerk
- No real model calls in the pipeline, no GitHub writes, no Search Console data
- Template uploads and keys are stored on the local disk, localhost only

Backend build plan

Ten handoffs, each self-contained for a builder agent. The UI keeps the same hooks, so screens don't change when the backend lands.

PHASE	WORK	SIZE
B0	Accounts, environments, keys	S
B1	Schema, tenancy and access checks, Clerk organisations, members, presence, locks, audit	L
B2	Post lifecycle as durable workflows; autopost ship or hold; schedules	L
B3	Real agents: Anthropic gateway, spend-capped inference, prompts from skills, Langfuse traces	L
B4	Visual agent and assets: connections, encrypted secrets, R2 storage	L
B5	Template parsing and rendering on the server	M
B6	GitHub App, webhooks, auto-merge; GitLab	L
B7	Search Console import and KAS	M
B8	Spend ledger and billing (Autumn, Polar)	M
B9	Security review, hardening and pilot	M

Decisions needed

D1 · Own Convex deployment

Recommended: yes. Its tenancy and data differ from the existing app. Needed before B1.

D2 · Anthropic direct

Recommended: call Anthropic directly rather than through OpenRouter, which stays as a fallback. Needed before B3.

D3 · Bill per site or per workspace

The stack says per site; the multi-site model puts plan and seats on the workspace. Suggested: workspace plan plus a per-site item. Needed before B8.

D14 · Image licensing policy

Which sourced and generated images are allowed, per provider. Needed before a paid pilot.

Also open: a signed eval-first spec for autonomy level L3 and the cost-per-post budget, and the cost model (D15).